

AP-E5 Four-Dimensional Background-Field Gate: Charged Two-Colour Lattice Action and the Gauge–Meson–Ghost–Fermion $B = 1$ Hessian

Route F research note

16 July 2026

Abstract

We formulate a four-dimensional Euclidean lattice regularisation for testing the charged two-colour Route-F candidate near a static $B = 1$ mesonic background. The gauge sector admits the Wilson action and its tree-level Symanzik extension; the colour-singlet effective sector contains a neutral $SU(2)$ meson and a charge-two complex diquark. Background gauge fixing produces the Faddeev–Popov operator. Two-flavour Wilson fermions with a chiral meson insertion give an exact even–odd Schur factorisation. We derive the complete *formal* graded Hessian and its one-loop determinant curvature.

A deterministic 2^4 calculation then tests a strictly smaller frozen-background quadratic truncation. It verifies gauge/ghost ξ -dependence, meson and diquark proxy spectra, a Wilson–Dirac Schur complement, one collective fermion log-determinant curvature, a negative control, a $B = 1$ continuum extrapolation, and Wilson versus tree-Symanzik cutoff orders. This is not importance sampling, dynamical four-dimensional QC2D, or a determinant-positivity proof. The complete nonlinear Skyrme Jacobi operator, collective modes, continuum renormalisation, Finkelstein–Rubinstein constraint, and global stability remain open; every physics-promotion and Route-E portal flag remains false.

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1 Scope and logical levels

Two-colour QCD has pseudoreal colour representations and meson–diquark competition that cannot be imported from three-colour intuition [1, 2]. We separate:

- (L1) the exact definition of a lattice action and its formal second variation;
- (L2) a deterministic finite-background spectrum with no ensemble average;
- (L3) a volume- and continuum-controlled path-integral result.

Only the first two are addressed. Recent tensor-network work provides a different controlled regulator in one spatial dimension, not a substitute for this four-dimensional problem [3]. Genuine four-dimensional QC2D simulations on 40^4 and 32^4 lattices exist [4], but their matter content and observables do not include the extra charged-scalar sector or the coupled $B = 1$ Hessian required here.

Fail-closed gate 1.1 (Meaning of a green card). A green card certifies the discretisation, projectors, Schur identity, scaling tests, and negative control. It does not certify Monte Carlo, measure positivity, the interacting continuum phase, or quantum stability.

2 Four-dimensional gauge and ghost blocks

2.1 Wilson and Symanzik-compatible actions

On a periodic lattice Λ_a , let

$$\nabla_\mu^+ f_x = \frac{f_{x+\hat{\mu}} - f_x}{a}, \quad \nabla_\mu^- f_x = \frac{f_x - f_{x-\hat{\mu}}}{a}.$$

The provisional gauge group is $SU(2)_c \times U(1)_g$, with links $V_\mu(x) \in SU(2)_c$ and $z_\mu(x) = e^{iaqb_\mu(x)}$. The Wilson action is [5]

$$S_{g,W} = \beta_c \sum_P \left(1 - \frac{1}{2} \Re \text{Tr } V_P \right) + \beta_g \sum_P (1 - \Re z_P). \quad (1)$$

A tree-level Lüscher–Weisz/Symanzik-compatible extension is

$$S_{g,\text{imp}} = \beta \left[c_0 \sum_P \left(1 - \frac{1}{2} \Re \text{Tr } W_P \right) + c_1 \sum_R \left(1 - \frac{1}{2} \Re \text{Tr } W_R \right) \right], \quad c_0 + 8c_1 = 1, \quad (2)$$

with $c_1 = -1/12$ at tree level [6, 7]. The matrix test uses Wilson $c_1 = 0$; a separate free dispersion test checks the improved branch. This establishes compatibility, not an interacting improved simulation. Here W denotes either gauge factor in its own defining representation; Eq. (2) is not a mixed $SU(2)_c$ – $U(1)_g$ plaquette.

2.2 Quadratic gauge fixing and FP cancellation

Expanding $V_\mu = e^{iaA_\mu^a T^a}$ and adding

$$S_{\text{gf}} = \frac{1}{2\xi} \sum_x (\nabla_\mu^- A_\mu^a)^2$$

gives

$$K_{\mu\nu}^{ab}(p) = \delta^{ab} [\hat{p}^2 \delta_{\mu\nu} + (\xi^{-1} - 1) \hat{p}_\mu \hat{p}_\nu^*], \quad \hat{p}_\mu = \frac{e^{iap_\mu} - 1}{a}. \quad (3)$$

Every $p \neq 0$ has three transverse eigenvalues $\hat{p}^2$ and one longitudinal eigenvalue $\hat{p}^2/\xi$. Background covariant gauge has

$$M_{\text{FP}}^{ab} = -\overline{D}_\mu^{ac} D_\mu^{cb};$$

at the tested trivial gauge background it is

$$M_{\text{FP}}^{ab} = -\delta^{ab} \nabla_\mu^- \nabla_\mu^+. \quad (4)$$

Removing the constant ghost and four gauge torons, one generator obeys

$$\frac{1}{2} \log \det' K_A(\xi) - \log \det' M_{\text{FP}} = \frac{1}{2} \log \det' K_A(1) - \log \det' M_{\text{FP}} - \frac{V-1}{2} \log \xi. \quad (5)$$

The final term is background independent and drops out of normalised background differences. Individual gauge eigenvalues are not gauge independent.

3 Charged meson–diquark sector

3.1 Action and $B = 1$ embedding

Let $U_x \in SU(2)$ be neutral and $\Delta_x \in \mathbb{C}$ carry $q_\Delta = 2$. A compatible EFT is

$$\begin{aligned} S_{\chi\Delta} = & \frac{f^2 a^2}{4} \sum_{x,\mu} \text{Tr}[(U_{x+\hat{\mu}} - U_x)^\dagger (U_{x+\hat{\mu}} - U_x)] + S_{\text{Sk},a}[U] \\ & + \frac{f^2 m_\pi^2 a^4}{2} \sum_x (2 - \text{Tr} U_x) + a^2 \sum_{x,\mu} |z_\mu(x)^2 \Delta_{x+\hat{\mu}} - \Delta_x|^2 \\ & + a^4 \sum_x [(m_\Delta^2 + \chi \rho_B(x)) |\Delta_x|^2 + \lambda_\Delta |\Delta_x|^4]. \end{aligned} \quad (6)$$

$S_{\text{Sk},a}$ is an explicitly chosen lattice transcription of the Skyrme term [8]; different transcriptions have different finite- a Hessians.

The sampled static hedgehog is

$$\overline{U}(\mathbf{x}) = \cos F(r) \mathbf{1} + i \sin F(r) \hat{\mathbf{x}} \cdot \boldsymbol{\tau}, \quad F(r) = 2 \tan^{-1} \left[\frac{R}{r} e^{-r^2/(2R^2)} \right]. \quad (7)$$

It has $F(0) = \pi, F(\infty) = 0$ and orientation

$$B = -\frac{1}{2\pi^2} \int_{\mathbb{R}^3} \det(n, \partial_1 n, \partial_2 n, \partial_3 n) d^3x = +1, \quad n = (\cos F, \hat{\mathbf{x}} \sin F). \quad (8)$$

Topology is measured on open grids and extrapolated in a^2 . The 2^4 Hessian only samples the same analytic background; it does not pretend that two spatial sites resolve topology.

3.2 Exact formal versus tested meson Hessian

Choose tangent frames $E_x : \mathbb{R}^3 \rightarrow T_{n_x} S^3$ and write

$$n_x(\eta) = n_x + E_x \eta_x - \frac{1}{2} n_x |\eta_x|^2 + \mathcal{O}(\eta^3).$$

The exact nonlinear lattice Jacobi block is

$$(K_\pi)_{xi,yj} = \frac{\partial^2}{\partial \eta_x^i \partial \eta_y^j} (S_\chi + S_{\text{Sk},a})[n(\eta)] \Big|_{\eta=0}. \quad (9)$$

It contains connection, target-curvature, mass-curvature, and every Skyrme second variation. The finite certificate tests only the declared positive frozen-background Gauss–Newton block

$$S_{\pi, \text{test}}^{(2)} = \frac{1}{2} \sum_{\langle xy \rangle} |\eta_y - T_{yx} \eta_x|^2 + \frac{1}{2} \sum_x [m_\pi^2 + \kappa_\pi \rho_B(x)] |\eta_x|^2, \quad (10)$$

where $T_{yx} \in SO(3)$ is Levi–Civita parallel transport on S^3 . Proxy positivity is therefore not reported as positivity of Eq. (9).

3.3 Charge-two diquark Hessian

At $\bar{\Delta} = 0$, the two real components share

$$K_\Delta = -D_\mu^{(2)-} D_\mu^{(2)+} + m_\Delta^2 + \chi \rho_B(x). \quad (11)$$

Because $D_\mu \Delta$ is linear in b_μ and Δ , $K_{b\Delta}|_{\bar{\Delta}=0} = 0$. A condensed phase instead has gauge–Goldstone mixing and requires an R_ξ treatment.

4 Wilson fermions and Schur reduction

4.1 Two-flavour operator

For colour doublets and two flavours, use [9]

$$D_{xy} = [m_0 + 4r/a + y_s \sigma_x + i y_p \gamma_5 \boldsymbol{\tau} \cdot \boldsymbol{\pi}_x] \delta_{xy} - \frac{1}{2a} \sum_\mu [(r - \gamma_\mu) W_\mu(x) \delta_{x+\hat{\mu},y} + (r + \gamma_\mu) W_\mu^\dagger(x - \hat{\mu}) \delta_{x-\hat{\mu},y}]. \quad (12)$$

The tested matrix sets colour and Abelian links to identity. Its 4 spin times 2 flavour components already contain $N_f = 2$; there is no extra flavour multiplier in the log determinant. One 128×128 spin–flavour matrix is diagonalised, and the two trivial colour copies are recorded as a degeneracy. This is not a measure-positivity argument.

A standard Wilson kernel can seed an overlap construction [10], but only after an admissibility/domain-wall gap and index analysis. Neither follows from a single finite singular value.

4.2 Exact even–odd identity

Even lattice extents make hopping bipartite:

$$D = \begin{pmatrix} D_{ee} & D_{eo} \\ D_{oe} & D_{oo} \end{pmatrix}, \quad S_e = D_{ee} - D_{eo} D_{oo}^{-1} D_{oe}. \quad (13)$$

Block elimination gives

$$\det D = \det D_{oo} \det S_e, \quad (14)$$

$$\begin{aligned} S_e x_e &= b_e - D_{eo} D_{oo}^{-1} b_o, \\ x_o &= D_{oo}^{-1} (b_o - D_{oe} x_e). \end{aligned} \quad (15)$$

Both complex determinant phase and the solution are checked against the unreduced matrix.

5 Formal graded Hessian and one-loop curvature

Let $\Phi = (A_\mu^a, b_\mu, \eta^i, d_1, d_2)$ and expand about $(0, 0, \bar{U}, 0, 0)$. Since a bosonic fluctuation times a fermion bilinear is cubic at $\bar{\psi} = 0$, the classical graded Hessian is

$$\mathcal{H}_{\text{cl}} = \left(\begin{array}{ccccc|cc} K_{AA} & K_{Ab} & K_{A\pi} & K_{Ad_1} & K_{Ad_2} & 0 & 0 \\ K_{bA} & K_{bb} & K_{b\pi} & K_{bd_1} & K_{bd_2} & 0 & 0 \\ K_{\pi A} & K_{\pi b} & K_{\pi\pi} & K_{\pi d_1} & K_{\pi d_2} & 0 & 0 \\ K_{d_1 A} & K_{d_1 b} & K_{d_1 \pi} & K_{\Delta} & 0 & 0 & 0 \\ K_{d_2 A} & K_{d_2 b} & K_{d_2 \pi} & 0 & K_{\Delta} & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & D \\ 0 & 0 & 0 & 0 & 0 & -D^T & 0 \end{array} \right), \quad (16)$$

plus $\bar{c}M_{\text{FPC}}$. In the tested neutral-meson, $\bar{\Delta} = 0$, trivial-gauge background, bosonic off-diagonal blocks vanish at tree level. Equation (16) is the complete *formal* block structure; the numerical operator substitutes Eq. (10) for the exact nonlinear K_π .

At one loop,

$$\Gamma^{(1)} = \frac{1}{2} \log \det' K_B - \log \det' M_{\text{FP}} - \Re \log \det D_{N_f=2}. \quad (17)$$

For coordinates q_i ,

$$\partial_i \partial_j \Gamma_f^{(1)} = \Re \text{Tr}[D^{-1} D_i D^{-1} D_j - D^{-1} D_{ij}]. \quad (18)$$

The trace includes both flavour components and, in the formal microscopic operator, both colours. The numerical trivial-colour-link test evaluates one colour copy; its full two-colour curvature is exactly twice the displayed one-copy result. For $D(t) = D + tQ$, the tested identity is

$$\Gamma_{f, \text{one colour}}''(0) = \Re \text{Tr}(D^{-1} Q D^{-1} Q), \quad (19)$$

not the full one-loop multi-field Hessian.

6 Zero modes and global obstructions

- (Z1) Constant ghosts and gauge torons are removed by primed determinants.
- (Z2) An exact infinite-volume soliton has translations and isorotations. Finite volume, anisotropy, and pinning lift them; the positive test block cannot count them.
- (Z3) A nonzero Wilson singular value is not an index. Spectral flow or an admissible overlap construction is required.
- (Z4) A nonnegative local Hessian does not exclude finite-amplitude unwinding. Coarse lattices do not preserve topology without a controlled admissibility or continuum argument.

The Finkelstein–Rubinstein sign concerns the fundamental group of physical configuration space and its quantisation, not an eigenvalue [11].

7 Deterministic numerical certificate

The finite benchmark uses a periodic 2^4 lattice, $L = 2.4$, $R = 1.2$, $m_\pi^2 = 0.25$, $\kappa_\pi = 0.18$, $m_\Delta^2 = 0.81$, $\chi = -0.30$, $r = 1$, $m_0 = 0.65$, $y_s = 0.20$, and $y_p = 0.16$. It is deliberately small so every determinant is directly reproducible.

7.1 Topology and bosonic spectra

The open-grid estimator uses the larger topology box $L_B = 6.4$ at $N_s = 17, 25, 33, 49, 65$, fitted quadratically in a^2 using the last four points, and gives

$$B(0) = 0.999981468235.$$

The 2^4 blocks give

$$\begin{aligned}\lambda_{\min}(K_{\pi, \text{test}}) &= 0.621297848377, \\ \lambda_{\min}(K_{\Delta}) &= 0.694069172619, \\ \lambda_{\min}(K_{\Delta}^{\text{negative control}}) &= -0.349308273811.\end{aligned}\tag{20}$$

The control changes only $\chi : -0.30 \mapsto -3.00$; its negative mode proves the sign audit can fail.

7.2 Gauge, fermion, and scaling residuals

The maximum corrected residual in Eq. (5) for $\xi = 0.5, 0.7, 1, 2$ is 1.42×10^{-14} . The Wilson matrix has

$$\sigma_{\min}(D) = 0.702570765924.$$

The determinant-log, determinant-phase, and Schur-solve residuals are

$$4.26 \times 10^{-14}, \quad 5.19 \times 10^{-17}, \quad 6.30 \times 10^{-16}.$$

The trace/finite-difference relative discrepancy in Eq. (19) is 1.32×10^{-8} .

At fixed physical L ,

$$\begin{aligned}\hat{p}_W^2 &= \frac{4}{a^2} \sin^2 \frac{ap}{2} = p^2 - \frac{a^2 p^4}{12} + \mathcal{O}(a^4), \\ \hat{p}_{\text{imp}}^2 &= \hat{p}_W^2 + \frac{a^2}{12} (\hat{p}_W^2)^2 = p^2 - \frac{a^4 p^6}{90} + \mathcal{O}(a^6).\end{aligned}\tag{21}$$

The $N = 32, 64$ observed orders are

$$p_W = 1.99860957, \quad p_{\text{imp}} = 3.99627578.$$

This tests free quadratic scaling, not interacting renormalisation.

8 Status table and exact fail-closed keys

Claim	Status	Reason
Wilson/Symanzik-compatible action	defined	Equations (1)–(6).
Gauge/ghost projectors and ξ law	finite pass	Prime spectra test Eq. (5).
Wilson even-odd Schur algebra	finite pass	Determinant and solve identities agree.
Complete nonlinear $B = 1$ Hessian	open	The test uses Eq. (10), not Eq. (9).
Dynamical four-dimensional QC2D phase	open	No ensemble, importance sampling, or thermodynamic limit.
Positive fermion measure	open	A singular value is not an antiunitary/global-form proof.

Claim	Status	Reason
FR and finite-amplitude stability	open	Require configuration-space topology and global minimax analysis.
Degree-one Route-E portal	blocked	Preceding microscopic and global gates remain false.

Fail-closed gate 8.1 (Machine-readable promotion boundary). The JSON card sets each of the following exact keys to `false`:

- `importance_sampling_or_Monte_Carlo_performed`
- `dynamical_4d_lattice_QC2D_phase_established`
- `fermion_determinant_positivity_proven`
- `complete_nonlinear_gauge_meson_ghost_fermion_Hessian_computed`
- `Skyrme_Jacobi_translation_isorotation_zero_modes_resolved`
- `one_loop_renormalised_continuum_limit_established`
- `Finkelstein_Rubinstein_constraint_derived`
- `finite_amplitude_global_stability_established`
- `degree_one_route_E_portal_allowed`
- `physics_promotion_allowed`

9 Three routes forward

- (R1) **Full classical Jacobi route.** Choose one explicit lattice Skyrme transcription, relax the $B = 1$ background at each a, L , assemble its exact sparse second variation, project gauge, translation, and isorotation directions, and extrapolate all low modes.
- (R2) **Dynamical QC2D route.** First prove the antiunitary symmetry and measure sign for the actual charges and global form. Then use HMC or an independently validated alternative, multiple spacings and volumes, scale setting, and correlator diquark gaps.
- (R3) **Fermion/index route.** Establish link admissibility, compute overlap spectral flow in the relaxed $B = 1$ background, determine the Callias/APS determinant line, and combine its curvature with the projected bosonic Jacobi operator and FR quantisation.

10 Reproducibility

The deterministic verifier is `route_f/code/verify_ap_e5_4d_qc2d_lattice_hessian.py`. It writes JSON and Markdown cards, records source hashes and library versions, uses seed 20260716 only for a Schur-solve right-hand side, and exits nonzero if any declared check fails.

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